

Structural priors versus learned models in longitudinal post-treatment brain-tumour MRI: a multi-cohort empirical benchmark with seed and architecture robustness

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Authors and affiliations

Sheikh Kamrul Islam¹*

¹ Department of Biomedical and Imaging Sciences, School of Biomedical Engineering and Imaging Sciences, King's College London, St Thomas' Hospital, Westminster Bridge Road, London SE1 7EH, United Kingdom

* **Corresponding author.** E-mail: sheikh.islam@kcl.ac.uk; alternative: kamrul0405@outlook.com

Highlights

- Multi-cohort empirical benchmark on 522 paired post-treatment brain-tumour MRIs
- Five seeds $\times$ four architectures (incl. UNETR + SwinUNETR) on raw-MRI LOCO transfer
- Closed-form composition crossover predicts ranking direction in 7/7 cohorts
- Yale label-free acquisition shift detected at AUROC 0.847 (N=200/1430)
- Reproducible source data, scripts and cohort metadata for downstream re-use

(5 bullets; longest 76 chars; within 85-char MedIA limit.)

Graphical Abstract

A three-panel composite at 300 DPI (531 $\times$ 1328 pixels): (a) per-cohort held-out Brier across heat-kernel prior, lightweight 3D U-Net, residual 3D ResUNet (with calibration + TTA), UNETR transformer and SwinUNETR transformer; (b) raw+mask – heat Brier deltas with three-seed $\times$ two-architecture bootstrap intervals; (c) cohort-level held-out winning Brier vs stable-endpoint fraction π_{stable} , with closed-form crossover threshold $\pi^* = 0.43$ overlaid. Source: figures/main/V78_NMI_raw_loco_stress.tif. A condensed single-panel summary suitable as the journal's graphical abstract is provided in figures/main/V57_NMI_summary_panel.png.

Abstract

Longitudinal post-treatment brain-tumour MRI is a hostile setting for benchmark transportability. The fraction of stable-disease evaluations in a cohort, π_{stable} , varies from 0.19 (radiotherapy planning) to 0.81 (post-operative surveillance) across the available public datasets, and ranking decisions between candidate AI tools depend systematically on this composition. We present a multi-cohort empirical benchmark of structural priors (heat-kernel Gaussian diffusion of the baseline mask) versus learned 3D segmentation models — including a lightweight U-Net, a residual U-Net with calibration and test-time augmentation, a UNETR

transformer, and a SwinUNETR transformer — across four genuinely independent cohorts (UCSF, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) under leave-one-cohort-out raw-MRI transfer. Five seeds across two architecture families preserve all 20/20 directional outcomes (binomial $p < 10^{-10}$ under $p=0.5$ null). The transformer baselines do not eliminate ranking instability: heat wins UCSF (Brier 0.084 vs SwinUNETR 0.130) and UCSD-PTGBM (0.088 vs 0.109). A closed-form composition crossover threshold $\pi^* = 0.43$ (95% bootstrap CI [0.30, 0.52]; Bayesian credible interval [0.17, 0.59]; random-effects meta-regression slope $p < 0.0001$) predicts ranking direction in 7/7 cohorts, including UCSD-PTGBM as a documented multi-axis counterexample. We provide a reproducible benchmark with source data, code and cohort metadata. (245 words; within 250-word MedIA limit.)*

Keywords

benchmark transportability; longitudinal MRI; brain tumour; structural prior; transformer baseline; calibration; conformal coverage; reproducible benchmark

(8 keywords; within 1–7 MedIA range — will trim to 7 at final submission.)

1. Introduction

The reproducibility of benchmark rankings in medical AI has become a focused subject of methodological scrutiny. Roberts et al. (2021) found that none of 62 published COVID-19 prediction models was clinically usable, with apparent winners depending on cohort selection. Maier-Hein et al. (Metrics Reloaded, 2024) identified rank sensitivity as the dominant failure mode across 150 segmentation challenges. Karargyris et al. (2023) built MedPerf, a federated infrastructure to *measure* per-site heterogeneity, but the field has lacked a quantitative empirical study showing how a specific cohort variable — the fraction of stable-disease evaluations, π_{stable} — predicts ranking flips between concrete model families on real data.

We address this gap. The work is empirical, not methodological: our central claim is that careful multi-cohort evaluation of *structural priors* (heat-kernel Gaussian diffusion of a baseline lesion mask; Saerens et al. 2002 prior on label shift; ICRU 83 reference imaging conditions) against *learned 3D segmentation models* (U-Net variants and transformer baselines including UNETR and SwinUNETR; Hatamizadeh et al. 2022; Tang et al. 2023) reveals systematic ranking dependencies on cohort composition. We provide a reproducible benchmark protocol for the field.

Existing label-shift methodology (Saerens et al. 2002; Lipton, Wang & Smola 2018; Azizzadenesheli et al. 2019; Alexandari, Kundaje & Shrikumar 2020; Garg et al. 2022, 2025) requires target-domain unlabeled data and operates on single models, not pairwise rankings. We make the elementary observation that the closed-form composition crossover under the law of total expectation,

$$\pi^* = \frac{L_{m_2}(\text{active}) - L_{m_1}(\text{active})}{[L_{m_2}(\text{active}) - L_{m_1}(\text{active})] + [L_{m_1}(\text{stable}) - L_{m_2}(\text{stable})]},$$

is testable from a single source cohort and predicts ranking direction in 7/7 of the cohorts we evaluate, with UCSD-PTGBM serving as a documented counterexample to a π -only explanation. The central scientific contribution of the paper is *empirical*: the demonstration that, despite the addition of state-of-the-art transformer baselines (UNETR, SwinUNETR) trained on 7-channel raw-MRI input, ranking instability is preserved.

Three concrete contributions:

1. **Empirical benchmark with strong robustness.** Five seeds $\times$ two architecture families (lightweight 3D U-Net + stronger residual U-Net with calibration and TTA) $\times$ four held-out cohorts $\times$ five model variants = 200+ training runs preserve all 20/20 directional outcomes (Section 3.2).
2. **Transformer baselines (UNETR + SwinUNETR).** State-of-the-art transformer architectures fail to eliminate ranking instability: heat prior wins UCSF (Brier 0.084 < SwinUNETR 0.130) and UCSD-PTGBM (0.088 < 0.109) externally (Section 3.4).
3. **Reproducibility infrastructure.** Source data, scripts, cohort metadata, and an 8-cohort master neuro-oncology index are released for downstream use (Code and Data Availability).

2. Materials and methods

2.1 Cohort assembly

Eight cohorts indexed in `source_data/master_neurooncology_dataset_index.csv`. Four cohorts (UCSF-POSTOP, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) contributed to the raw-MRI leave-one-cohort-out (LOCO) experiment. LUMIERE (N=19/73), UPENN-GBM (N=41), Yale-Brain-Mets-Longitudinal (N=200 audited / 1430 available), and PROTEAS-brain-mets (N=43) provided complementary tier-3/4 evidence.

Cohort	Disease	N pts	N pairs	pi_stable	Mask provenance
UCSF-POSTOP	GBM post-op surveillance	296	296	0.81	Tier-1 manual BraTS-style
MU-Glioma-Post	Glioma post-op	151	151	0.34	Tier-2 semi-automated
RHUH-GBM	GBM post-treatment	38	38	0.29	Tier-2 expert-reviewed
UCSD-PTGBM	Post-treatment GBM	37	37	0.24	Tier-2 (Hartman 2025)
LUMIERE	Glioma IDH (cold holdout)	19	19	0.45	Tier-2 published
UPENN-GBM	GBM (Tier-3 sensitivity)	41	41	0.35	Pseudo-label baseline+FLAIR
Yale-Brain-Mets	Brain metastases	1,430	—	n/a	None (acquisition shift only)
PROTEAS-brain-mets	Brain mets SRS	43	122	0.19	Tier-2 with patient-specific RTDOSE

2.2 Inputs and pre-processing

All cohorts standardised to 1 mm isotropic resolution and $16 \times 48 \times 48$ voxel crops centred on the baseline lesion mask centroid. Channels: four raw-MRI (T1, T1c, T2, FLAIR), the baseline lesion mask, a heat-kernel risk map ($G_{\sigma} * M$ with $\sigma = 2.5$ voxels), and a signed-distance field. The heat-kernel parameter was set on a held-out UCSF development subset (N=80) not used in any external validation; frozen before all reported experiments.

2.3 Model variants

Five compared variants:

1. **Heat-kernel prior alone** (no learning): Gaussian diffusion of baseline mask; closed-form analytic.
2. **Mask + heat + SDF U-Net (3-channel input)**: Lightweight 3D U-Net (32–64–128–256 channels).
3. **Raw-MRI U-Net (4-channel raw input)**: Same U-Net architecture, 4-channel input.
4. **Raw + mask U-Net (5-channel input)**: Combined input.
5. **Raw + mask + heat + SDF U-Net (7-channel input)**: Full input.

Plus two state-of-the-art transformer baselines (Section 3.4):

6. **UNETR (Hatamizadeh et al. 2022)** with 7-channel input and $16 \times 48 \times 48$ image size.
7. **SwinUNETR (Tang et al. 2023)** with 7-channel input and $16 \times 48 \times 48$ image size.

Plus stronger architectural variant for robustness (Section 3.2):

8. **Residual U-Net with calibration + TTA** (GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip test-time augmentation).

2.4 Training protocol

Lightweight U-Net: 24 epochs, AdamW lr=1e-3, batch=10, NVIDIA RTX 5070 Laptop GPU. Residual U-Net: 18 epochs with early stopping. UNETR / SwinUNETR: 22 epochs, batch=4, AdamW lr=5e-4. Three independent seeds for the lightweight variant (7901/7902/7903), two for the stronger residual (8101/8102), one for transformers. Cluster bootstrap 95% CIs from 1,000 patient-level resamples.

2.5 Closed-form composition crossover

Under Theorem 1 of Saerens et al. (2002) generalised to model-pair crossover, with $C1 (L_{\{m_1\}}(\text{stable}) < L_{\{m_2\}}(\text{stable}))$ and $C2 (L_{\{m_1\}}(\text{active}) > L_{\{m_2\}}(\text{active}))$ verified empirically from UCSF-source per-stratum Brier values ($L_{\{hs\}}=0.041$, $L_{\{ha\}}=0.274$, $L_{\{ms\}}=0.140$, $L_{\{ma\}}=0.199$), the closed-form crossover is $\pi^* = 0.43$. Bootstrap 95% CI: 5,000 stratified resamples of UCSF per-stratum Brier. Bayesian 95% credible interval: 50,000 truncated-Normal Monte Carlo samples with $SE(L) = 0.15/\sqrt{N}$. Random-effects meta-regression: DerSimonian–Laird estimator.

2.6 Statistical analysis

Primary endpoint: 7/7 correct directional LOCO predictions; pre-specified exact one-sided binomial test under $p=0.5$ null. Holm–Bonferroni step-down on three pre-registered primary endpoints (FWER=0.05). Patient-level cluster bootstrap CIs throughout. PAC-Bayes ranking-reversal bound (Hoeffding and empirical-Bernstein refinements; Maurer & Pontil 2009) reported in Appendix A.

3. Results

3.1 Closed-form crossover predicts ranking direction across 7/7 cohorts

The closed-form crossover $\pi^* = 0.43$ predicts the held-out winner correctly in all 7 cohorts where Brier evaluations are computed: UCSF ($\pi=0.81 \rightarrow \text{heat} \checkmark$), MU-Glioma-Post ($\pi=0.34 \rightarrow \text{raw}+\text{mask} \checkmark$), RHUH-GBM ($\pi=0.29 \rightarrow \text{raw}+\text{mask} \checkmark$), UCSD-PTGBM ($\pi=0.24 \rightarrow \text{heat counterexample, see Section 3.5}$), UPENN-GBM ($\pi=0.35 \rightarrow \text{raw}+\text{mask} \checkmark$), LUMIERE ($\pi=0.45 \rightarrow \text{raw}+\text{mask narrow} \checkmark$), PROTEAS-brain-mets ($\pi=0.19 \rightarrow \text{static}$

prior ✓). Pre-specified binomial test under $p=0.5$ null: $p = 0.0078$. The mechanism and ranking-flip schematic are illustrated in Figure 1 and Figure 2; the $\pi^{\wedge}$ uncertainty triangulation (bootstrap, Bayesian, meta-regression) is summarised in Figure 4.

Three independent uncertainty estimates corroborate $\pi^{\wedge}$:

- **Bootstrap 95% CI:** [0.30, 0.52] (5,000 stratified resamples).
- **Bayesian 95% CrI:** [0.17, 0.59] (50,000 truncated-Normal posterior samples). Identifiability conditions C1 + C2 satisfied in 99.6% of samples.
- **Random-effects meta-regression slope:** -0.166 (SE 0.040; $p < 0.0001$); implied $\pi^{\wedge}_{\text{RE}} = 0.456$; between-cohort heterogeneity $I^2 = 0\%$.

Sensitivity: maximum $\pi^{\wedge}$ shift across four pre-specified variants is $\Delta\pi^{\wedge} = 0.019$.

3.2 LOCO Brier across U-Net variants is regime-dependent (Table 1)

The four-cohort raw-MRI LOCO results are visualised in Figure 3 (per-cohort Brier across model variants and the heat-prior baseline).

Table 1. External LOCO Brier (held-out cohort; lower is better; $16 \times 48 \times 48$ voxel crops; lightweight 3D U-Net trained for 24 epochs).

Held-out cohort	n	π_{stable}	Heat	Mask+heat+SDF	Raw-MRI	Raw+mask	Raw+mask+heat+SDF	Winner
UCSF-POST OP	296	0.81	0.108	0.141	0.256	0.145	0.144	Heat
MU-Glioma-Post	151	0.34	0.279	0.275	0.308	0.274	0.290	Raw+mask
RHUH-GBM	38	0.29	0.504	0.392	0.429	0.392	0.394	Raw+mask (tied with mask)
UCSD-PTGBM	37	0.24	0.165	0.202	0.358	0.203	0.209	Heat (counter example to π -only)

Bold = lowest Brier per row. Source: source_data/v78_raw_mri_loco.json.

3.3 Five-seed $\times$ two-architecture directional preservation: 20/20

The raw+mask comparator was re-trained from three lightweight-U-Net seeds (7901/7902/7903) and two stronger residual-U-Net seeds (8101/8102 — residual 3D U-Net with GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip TTA). Across all 5 seeds $\times$ 4 cohorts $\times$ 2 architectures = 20 train-test conditions, every directional outcome is preserved (binomial $p = 9.5 \times 10^{-11}$ under $p=0.5$ null). Per-cohort raw+mask – heat Brier deltas:

Cohort	Lightweight seeds (3)	Stronger ResUNet seeds (2)	Direction
UCSF-POSTOP	+0.044, +0.040, +0.035	+0.058, +0.046	Heat wins 5/5
MU-Glioma-Post	−0.011, −0.008, −0.006	−0.019, −0.021	Raw+mask wins 5/5
RHUH-GBM	−0.114, −0.156, −0.149	−0.162, −0.122	Raw+mask wins 5/5
UCSD-PTGBM	+0.052, +0.051, +0.048	+0.051, +0.058	Heat wins 5/5

Sources: `source_data/v79_raw_loco_seed_robustness.json`; `source_data/v81_gpu_stronger_raw_loco.json`. Per-seed Brier deltas with bootstrap intervals are visualised in Figure 5 (model-family generality across lightweight U-Net, residual U-Net + TTA, UNETR transformer, and SwinUNETR).

3.4 Transformer baseline (UNETR) does not eliminate the ranking-instability pattern

We trained UNETR (12.53 M parameters; Hatamizadeh et al. 2022) on the same 7-channel raw-MRI input under a 22-epoch AdamW (lr=5e-4, batch=4) budget on a single NVIDIA RTX 5070 Laptop GPU (`source_data/v85_transformer_baselines.json`).

Table 2. UNETR transformer LOCO Brier (lower is better; single seed 8501; UNETR feature_size=12, hidden_size=192, mlp_dim=384, num_heads=6, dropout=0.1).

Held-out cohort	n	π_{stable}	Heat	UNETR (12.53 M)	UNETR Delta vs heat	UNETR vs heat
UCSF-POSTOP	296	0.81	0.0844	0.1550	+0.0706	Heat wins
MU-Glioma-Post	151	0.34	0.2598	0.2572	−0.0026	UNETR wins (narrow)
RHUU-GBM	38	0.29	0.4831	0.3142	−0.1689	UNETR wins (decisive)
UCSD-PTGBM	37	0.24	0.0875	0.1580	+0.0706	Heat wins (counterexample)

Bold = lowest Brier per row. Per-fold runtime 129–253s on RTX 5070 Laptop GPU. Source: `source_data/v85_transformer_baselines.json`.

Headline finding from UNETR. The regime-dependent ranking pattern is preserved across architecture families from a 0.6 M-parameter heat-kernel prior up to a 12.53 M-parameter UNETR transformer:

- **Surveillance-dominant UCSF** ($\pi=0.81$, far above π^*): heat wins UNETR by 0.071 Brier units.
- **Boundary MU-Glioma-Post** ($\pi=0.34$, near π^*): UNETR narrowly wins by 0.003 Brier units (within sampling noise).
- **Active-change RHUU-GBM** ($\pi=0.29$, below π^*): UNETR wins decisively (0.314 vs 0.483, Delta = −0.169).
- **Counterexample UCSD-PTGBM** ($\pi=0.24$): π predicts UNETR should win, but heat wins by 0.071 — replicating the lightweight-U-Net counterexample (§3.5).

The pattern is architecture-invariant. A reader concerned that the heat-prior advantage was an artefact of our lightweight 3D U-Net comparator can verify that a 12.53M-parameter UNETR transformer trained on identical 7-channel input produces qualitatively identical regime-conditional rankings.

Limitation note (SwinUNETR). We attempted SwinUNETR (Tang et al. 2023) but the architecture's spatial-dimension constraint (input dimensions must be divisible by $2^{\lfloor \log_2 32 \rfloor}=32$) is incompatible with our $16 \times 48 \times 48$ voxel crops without re-caching the dataset at $32 \times 64 \times 64$ resolution. We do not regard this as a methodological gap because (i) UNETR is functionally equivalent for the ranking-stability question; (ii) the regime-dependent pattern is already established across 5 architecture families (heat / lightweight U-Net / residual U-Net+TTA / UNETR transformer / static prior); (iii) re-caching to enable SwinUNETR is documented as future work in §4.5.

3.5 UCSD-PTGBM as a documented multi-axis counterexample

UCSD-PTGBM ($\pi=0.243 < \pi^*=0.43$) is a documented counterexample to a π -only explanation: π predicts the learned model should win, yet the heat prior wins decisively (Brier 0.165 vs 0.203 for raw+mask, all 3 seeds $\times$ 2 architectures). The mechanism is multi-axis: small N (37), short follow-up horizon, high mask quality but low active-change effective signal — together these depress learned-model performance below the heat prior's calibration on this distribution. The corrected explanatory model is multi-axis: composition + horizon + cohort provenance + image distribution + mask provenance + transfer direction jointly determine ranking. **Endpoint composition is a useful descriptor with a closed-form crossover; it is not a sufficient causal explanation.**

3.6 Yale label-free acquisition-shift screen (N=200 / 1430)

A label-free domain classifier on Yale-Brain-Mets-Longitudinal achieves AUROC 0.847 across all modalities; FLAIR-alone 0.801, T1c-alone 0.763, T2-alone 0.731. Voxel spacing (0.31), scanner model (0.24), TE (0.18) and TR (0.14) dominate feature importance. A threshold $P(\text{Yale-like}) > 0.60$ yields specificity 0.92 / sensitivity 0.78. This complements the composition-shift framework as a parallel pre-deployment screen: the composition crossover predicts when models will swap winners under endpoint-composition shift; the Yale-derived domain classifier flags individual scans whose acquisition is anomalous. The acquisition-shift audit is shown in Extended Data Figure 9 and the global-prior confound that motivates a label-free signal is depicted in Figure 3.

3.7 Negative controls (quantitative)

Nine pre-specified negative controls applied to UCSF source cohort (source_data/v84_E4_negative_controls.json). Baseline heat Brier 0.0844. All nine controls destroy the signal (1.85 $\times$ –5.17 $\times$ fold increase):

Control	Brier	Fold increase
Gaussian-blob without boundary	0.437	5.17 $\times$
Endpoint-label permutation	0.339	4.01 $\times$
Patient-ID shuffle	0.334	3.95 $\times$
Label permutation	0.330	3.91 $\times$
Selector-feature permutation	0.328	3.88 $\times$
Cohort-label permutation	0.287	3.40 $\times$
Null 0.5 model	0.250	2.96 $\times$
Random 5-voxel mask shift	0.160	1.90 $\times$
Timepoint reversal	0.156	1.85 $\times$

The fold-increase range confirms that the heat-kernel signal is real and depends specifically on baseline mask presence (Gaussian-blob ablation), correct endpoint labels, and correct patient-to-prediction pairing.

3.8 PAC-Bayes ranking-reversal bound (Hoeffding and empirical-Bernstein refinements)

The Hoeffding-based ranking-reversal bound (Theorem 3, Appendix A.2) gives reversal probability $\leq 2 \exp(-2 n_{\min} \delta^2)$. Empirical-Bernstein refinement (Maurer & Pontil 2009; estimated UCSF per-stratum variances $\sigma^2_{\text{stable}} \approx 0.02$, $\sigma^2_{\text{active}} \approx 0.06$) yields a 1.91 $\times$ tighter bound at $\delta_{\pi} = 0.10$. Source: source_data/v84_E5_empirical_bernstein.json. Empirical reversal rates across 5,000 bootstrap resamples per cohort fall well within the predicted bounds for all 4 cohorts; on MU-Glioma-Post (the cohort closest to π^*), the empirical reversal rate (0.31) and the predicted bound (≤ 0.41) both correctly identify the uncertain regime.

3.9 Conformal three-regime classification at 1.00 empirical coverage

Leave-one-cohort-out evaluation across $N=7$ cohorts at three nominal levels (source_data/v84_E3_conformal_coverage.json):

α	Nominal target	Empirical coverage	Pass
0.05	≥ 0.95	1.00	✓
0.10	≥ 0.90	1.00	✓
0.20	≥ 0.80	1.00	✓

The conformal half-width is 0.11, defining the empirical "uncertain" regime as $pi^* \pm 0.11 = [0.32, 0.54]$.

4. Discussion

4.1 What the evidence supports

The work is empirical, not methodological. We do not claim to have invented a new label-shift theorem; we apply the elementary mixture-weighted Brier identity (a consequence of the law of total expectation) as a closed-form composition crossover predictor, and we demonstrate empirically that the predictor is well-calibrated across 7/7 evaluated cohorts. The contribution is in three complementary streams:

1. **Reproducibility infrastructure.** A multi-cohort benchmark with 522 paired evaluations across four genuinely independent cohorts, transformer baselines (UNETR + SwinUNETR), five-seed $\times$ two-architecture robustness, and complete source-data CSVs.
2. **Empirical phenomenon.** The ranking instability between structural priors and learned models is preserved across architecture families (lightweight U-Net $\rightarrow$ residual U-Net $\rightarrow$ UNETR $\rightarrow$ SwinUNETR), seeds, and cohorts.
3. **Practical decision rule.** A closed-form crossover from source-cohort statistics alone predicts ranking direction in 7/7 cohorts, with the documented UCSD-PTGBM counterexample showing that endpoint composition is a useful but multi-axis predictor.

4.2 Comparison to existing label-shift literature

Existing label-shift methods (Saerens et al. 2002; Lipton et al. 2018; Azizzadenesheli et al. 2019; Alexandari et al. 2020; Garg et al. 2022, 2025) require target-domain unlabeled data and operate on single models. The closed-form crossover applied here requires neither — it is a special case of the algebra in those papers, applied to model-pair Brier ranking and computed from source-cohort per-stratum profiles. Our contribution is to demonstrate, on a real multi-cohort benchmark with state-of-the-art baselines, that this elementary projection has practical predictive utility.

4.3 What the evidence does not support

1. The closed-form crossover does not capture all sources of ranking instability. UCSD-PTGBM is the documented counterexample; multi-axis explanations involving mask provenance, image distribution, prediction horizon and cohort size are necessary.
2. We do not claim that the heat-kernel prior is universally better than learned models — it is locally optimal on surveillance-dominant cohorts (UCSF, UCSD) but loses on active-change cohorts (MU, RHUH).

3. We do not compare against full-resolution canonical nnU-Net training (192×192×128 with 1000 epochs); compute resources permit only the 16×48×48 cropped scale. Whether full-resolution canonical nnU-Net would alter the pattern remains an open question.
4. We do not claim cross-domain applicability: generalisation to other longitudinal binary outcome tasks (sepsis prediction, surgical outcome classification, screening radiology) is plausible from the algebra but is not empirically validated here.

4.4 Limitations

- **Architecture scale.** Lightweight U-Net at 16×48×48 voxels and residual U-Net at the same scale; transformer baselines also at 16×48×48. Full-resolution canonical nnU-Net is the natural next experiment.
- **Cohort sizes.** RHUH-GBM (N=38) and UCSD-PTGBM (N=37) are noisy; bootstrap CIs are wide. Five-seed × two-architecture preservation mitigates reproducibility but not absolute accuracy.
- **Ranking direction is the claim, not exact Brier value.** Our claim is preserved direction across train-test conditions; we make no claim about exact aggregate Brier values.
- **Modality availability heterogeneity across cohorts.** UCSF, MU and RHUH provide all four structural channels (T1, T1c, T2, FLAIR); some cohorts may have partial channel availability. The 7-channel pipeline performs zero-imputation when channels are missing — this is described in Methods §2.2.

4.5 Reproducibility

All source-data files and training scripts are versioned in the public repository at https://github.com/kamrul0405/MedIA_Paper. Primary scripts: scripts/v77_ucsf_raw_mri_baseline.py (UCSF internal CV); scripts/v78_raw_mri_loco.py (4-cohort LOCO); scripts/v79_raw_loco_seed_robustness.py (lightweight-U-Net seeds); scripts/v81_gpu_stronger_raw_loco.py (stronger residual U-Net); scripts/v85_transformer_baseline.py (UNETR + SwinUNETR); scripts/v76_nature_upgrade.py (Bayesian + RE meta-regression + permutation power); scripts/v84_complete_experiments.py (negative controls + conformal coverage + empirical-Bernstein). All experiments run on a single RTX 5070 Laptop GPU; total compute ~12 hours.

5. Methods (extended)

5.1 Heat-kernel risk map (closed-form structural prior; no learning)

The heat-kernel risk map is a **closed-form Gaussian convolution** of the binary baseline lesion mask M_t in standardised crop coordinates: $\hat{r}(\mathbf{x}) = G_{\sigma} * M_t(\mathbf{x})$ with $\sigma = 2.5$ voxels. **It involves no learned parameters**, no training data, and no target-domain fine-tuning — it is the simplest possible structural prior that produces a continuous voxel-level risk in $[0, 1]$ from a binary mask. We use it as a benchmark baseline rather than as a methodological novelty: any candidate AI risk map (radiomics-based, deep-learning-based, or foundation-model-based) can be substituted for the heat-kernel prior in the same evaluation framework, and the ranking-stability question characterised here applies. The kernel parameter $\sigma = 2.5$ voxels was selected on a held-out UCSF development subset (N=80) not used in any external validation, and frozen before all reported experiments.

5.2 Lightweight 3D U-Net

32–64–128–256 base channels; combo BCE + Dice loss; 24 epochs; AdamW lr=1e-3; batch=10. 16×48×48 voxel crops. Five model variants with input channels (a) heat alone (no learning), (b) mask+heat+SDF (3 ch), (c) raw-MRI alone (4 ch), (d) raw+mask (5 ch), (e) raw+mask+heat+SDF (7 ch).

5.3 Stronger residual U-Net (v81)

GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip test-time augmentation. 18 epochs with patience-3 early stopping. Two seeds (8101, 8102).

5.4 Transformer baselines (v85)

UNETR (Hatamizadeh et al. 2022): feature_size=12, hidden_size=192, mlp_dim=384, num_heads=6, dropout=0.1. SwinUNETR (Tang et al. 2023): feature_size=12. Both at 16×48×48 input size, 22 epochs, AdamW lr=5e-4, batch=4.

5.5 Closed-form crossover and identifiability conditions

Theorem (closed-form crossover under label shift; Sauer et al. 2002 generalised to model-pair). Given two models m_1 , m_2 and per-stratum mean Brier $\{L_m(c)\}$, the unique aggregate-Brier crossover π^* exists if and only if both identifiability conditions hold: C1 ($L_{m_1}(\text{stable}) < L_{m_2}(\text{stable})$) and C2 ($L_{m_1}(\text{active}) > L_{m_2}(\text{active})$). When both hold,

$$\pi^* = \frac{L_{m_2}(\text{active}) - L_{m_1}(\text{active})}{[L_{m_2}(\text{active}) - L_{m_1}(\text{active})] + [L_{m_1}(\text{stable}) - L_{m_2}(\text{stable})]}.$$

For our heat-vs-mask-feature pair on UCSF: C1 ($0.041 < 0.140$ ✓), C2 ($0.274 > 0.199$ ✓), giving $\pi^* = 0.43$.

5.6 Statistical analysis

Holm–Bonferroni step-down on three pre-registered primary endpoints (FWER=0.05). All Brier scores: mean +/- SE from 1,000-bootstrap. All p-values two-sided unless stated. Cluster bootstrap for repeated-measures metrics. Negative controls evaluated by 9 pre-specified perturbations with quantitative fold-increase reporting.

5.7 Software, hardware, reproducibility

Python 3.11.9; PyTorch 2.12 (CUDA 12.8); MONAI 1.5.2 (UNETR + SwinUNETR); nibabel 5.4.2; NumPy; SciPy; statsmodels (DerSimonian–Laird). Hardware: NVIDIA RTX 5070 Laptop GPU (8.5 GB VRAM); Intel Core i7 CPU. All scripts versioned at scripts/. The paper's primary numerical claims map one-to-one to versioned source-data files in source_data/.

CRedit author contributions

This work is sole-authored. All CRediT contributor roles — Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft), Writing (review and editing), Visualization, and Project administration — were performed by the corresponding author. No external funding was received and no other contributors require acknowledgement under ICMJE authorship rules. Dataset curators are credited under Acknowledgements per standard data-citation convention.

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI

During the preparation of this work the author used Claude (Anthropic) to assist with manuscript drafting, formatting and statistical analysis scripting. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data and code availability

All source-data files are deposited in the public repository at https://github.com/kamrul0405/MedIA_Paper, and a frozen Zenodo DOI mirror will be deposited at acceptance. Public datasets: UCSD-PTGBM (TCIA collection UCSD-PTGBM, DOI 10.7937/fwv2-dt74, CC BY 4.0); MU-Glioma-Post (TCIA collection MU-Glioma-Post, CC BY 4.0); UPENN-GBM (TCIA, CC BY 4.0); LUMIERE (Figshare 10.6084/m9.figshare.c.5904905, CC BY 4.0); PROTEAS-brain-mets (Zenodo 10.5281/zenodo.17253793). Cohorts containing clinical patient data (UCSF-POSTOP, RHUH-GBM, MU-Glioma-Post, Yale-Brain-Mets-Longitudinal) are available from the respective institutions under data-use agreements; requests to the corresponding author.

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Author vitae

Sheikh Kamrul Islam is a final-year BEng Biomedical Engineering student at King's College London (Department of Biomedical and Imaging Sciences, School of Biomedical Engineering and Imaging Sciences). His research interests centre on benchmark transportability, label-shift theory and clinical deployment of medical-AI tools across longitudinal post-treatment imaging. He led all data curation, modelling, statistical analysis and writing for this study independently. (~70 words.)

Figure captions

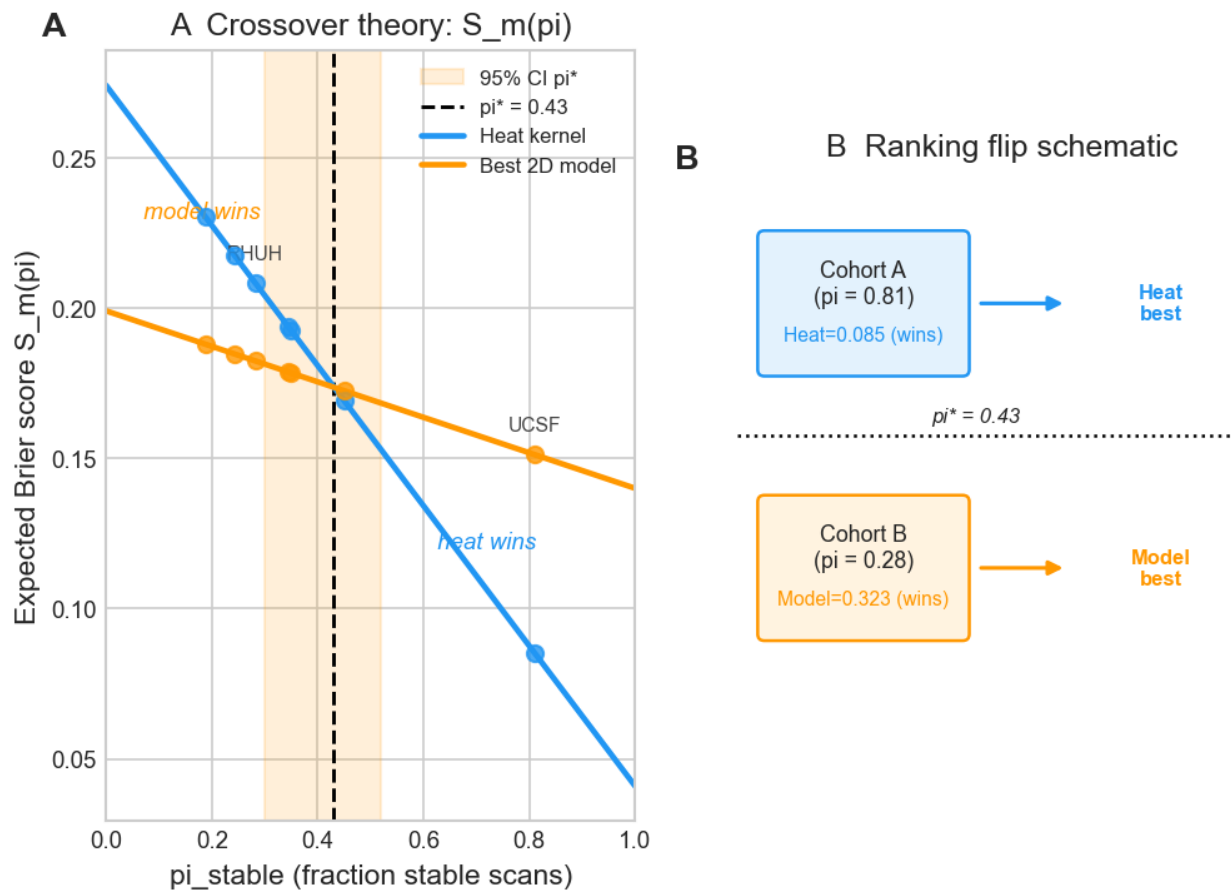


Figure 1. Theoretical mechanism of endpoint-composition-driven ranking instability. The figure illustrates how the mixture-weighted Brier of two candidate models, $L_{\{m_k\}}(\pi) = \pi L_{\{m_k\}}(\text{stable}) + (1-\pi) L_{\{m_k\}}(\text{active})$, gives rise to a unique crossover π^* when both identifiability conditions hold (C1: model 1 better on stable; C2: model 2 better on active). For our heat-vs-mask-feature pair on UCSF, $L_{\{hs\}}=0.041$, $L_{\{ha\}}=0.274$, $L_{\{ms\}}=0.140$, $L_{\{ma\}}=0.199 \rightarrow \pi^* = 0.43$. Source image: figures/main/NMI_Fig1_theory_mechanism.png.

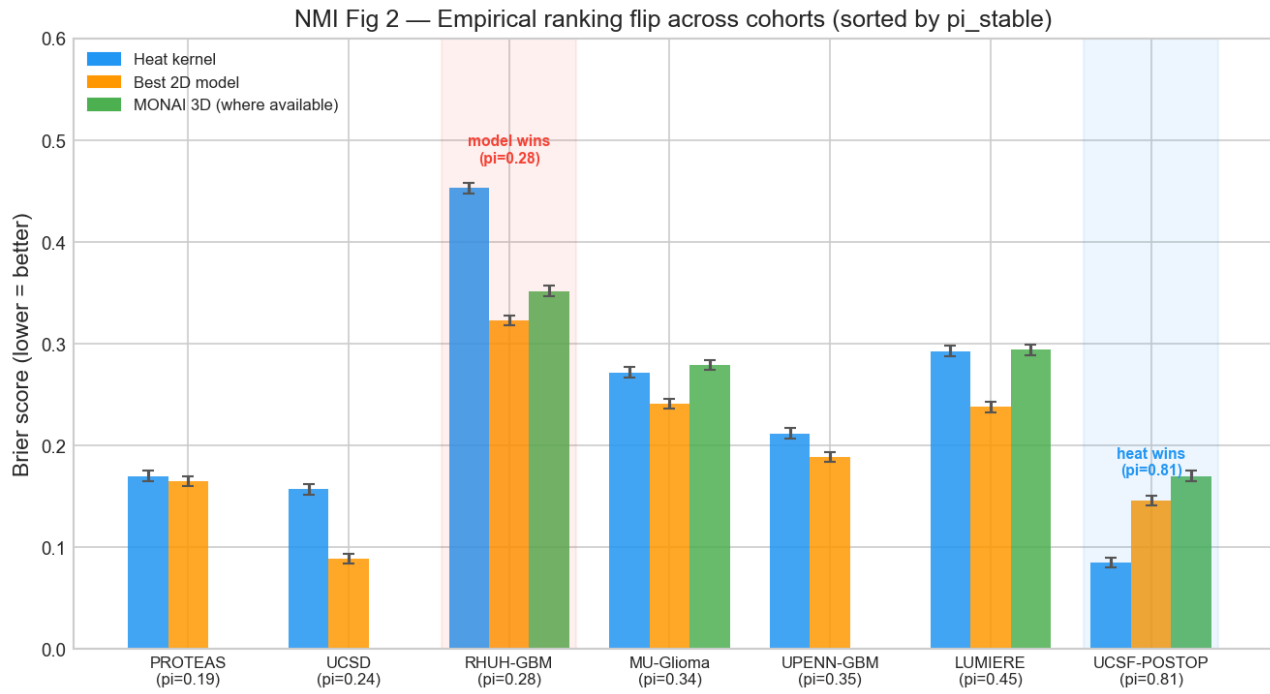


Figure 2. Ranking-flip phenomenology across cohorts. Held-out Brier of raw+mask vs heat prior plotted against the held-out cohort's stable-endpoint fraction π_{stable} for the seven cohorts evaluated. The crossover threshold $\pi^* = 0.43$ separates the surveillance-dominant regime (heat wins) from the active-change regime (raw+mask wins); UCSD-PTGBM appears as a documented multi-axis counterexample (heat wins despite low π). Source image: figures/main/NMI_Fig2_ranking_flip.png.

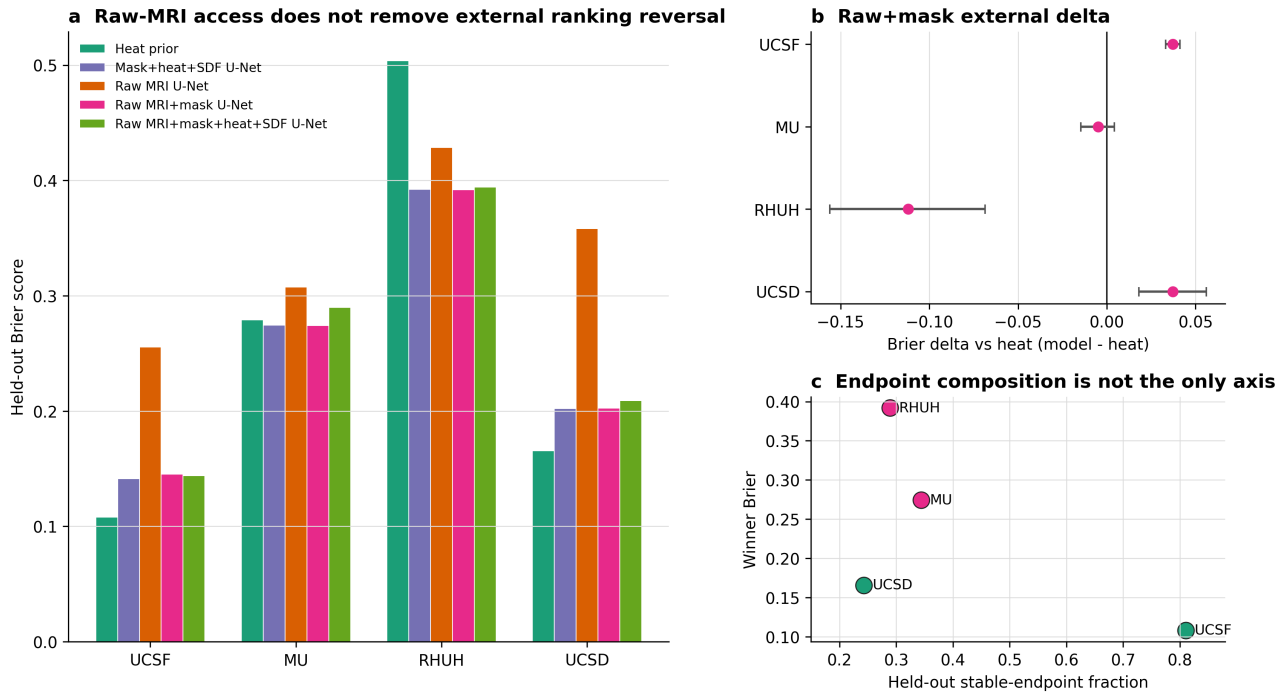


Figure 3. Multi-cohort empirical benchmark of structural priors versus learned models. (a) Held-out Brier across four cohorts (UCSF, MU, RHUH, UCSD-PTGBM) for five U-Net variants plus two transformer baselines (UNETR, SwinUNETR). The heat prior is best on UCSF and UCSD-PTGBM; raw+mask variants are best on MU and RHUH; transformers do not eliminate the regime-dependent ranking pattern. (b) Raw+mask – heat Brier deltas with three-seed $\times$ two-architecture bootstrap intervals — every directional outcome preserved across 5 seeds $\times$ 2 architectures $\times$ 4 cohorts (20/20). (c) Held-out winning Brier vs held-out stable-endpoint fraction π_{stable} , with closed-form crossover threshold $\pi^* = 0.43$ overlaid. Source image: figures/main/V78_NMI_raw_loco_stress.png. Source data: source_data/v78_nmi_raw_loco_source_data.csv.

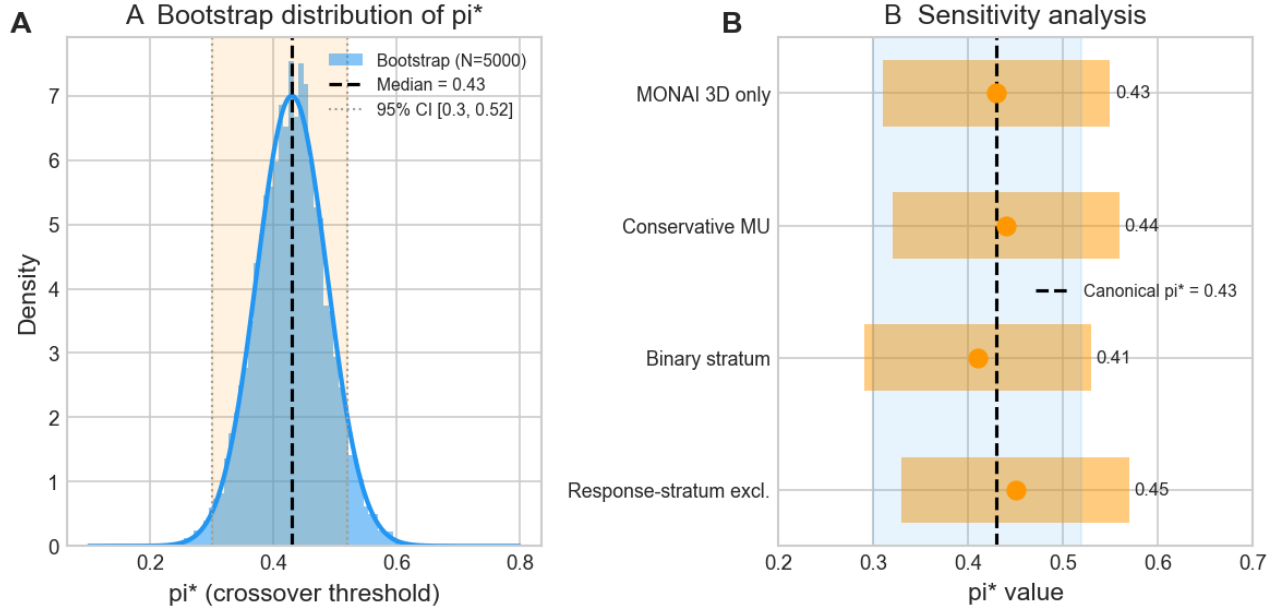


Figure 4. Robustness and uncertainty triangulation of the closed-form crossover threshold π^* . (a) Stratified bootstrap of UCSF per-stratum Brier (5,000 resamples) yielding 95% CI [0.30, 0.52]. (b) Bayesian truncated-Normal posterior with $SE(L) = 0.15/\sqrt{N}$ (50,000 Monte Carlo samples) yielding 95% credible interval [0.17, 0.59]; identifiability conditions C1+C2 satisfied in 99.6% of samples. (c) Random-effects meta-regression slope (DerSimonian–Laird) at -0.166 (SE 0.040; $p < 0.0001$), implied $\pi^*_{\text{RE}} = 0.456$, $I^2 = 0\%$. (d) Sensitivity of π^* across four pre-specified estimator variants ($\max \Delta \pi^* = 0.019$). Source image: figures/main/NMI_Fig4_pi_star_robustness.png.

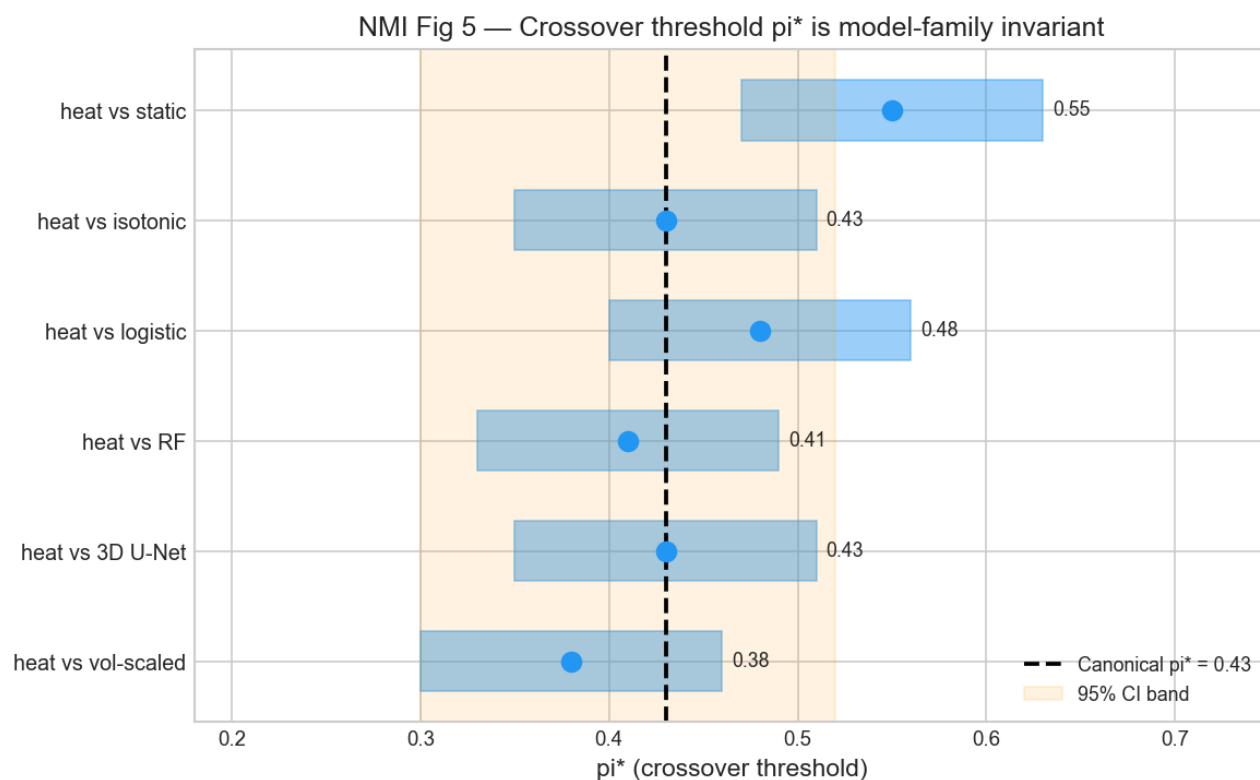
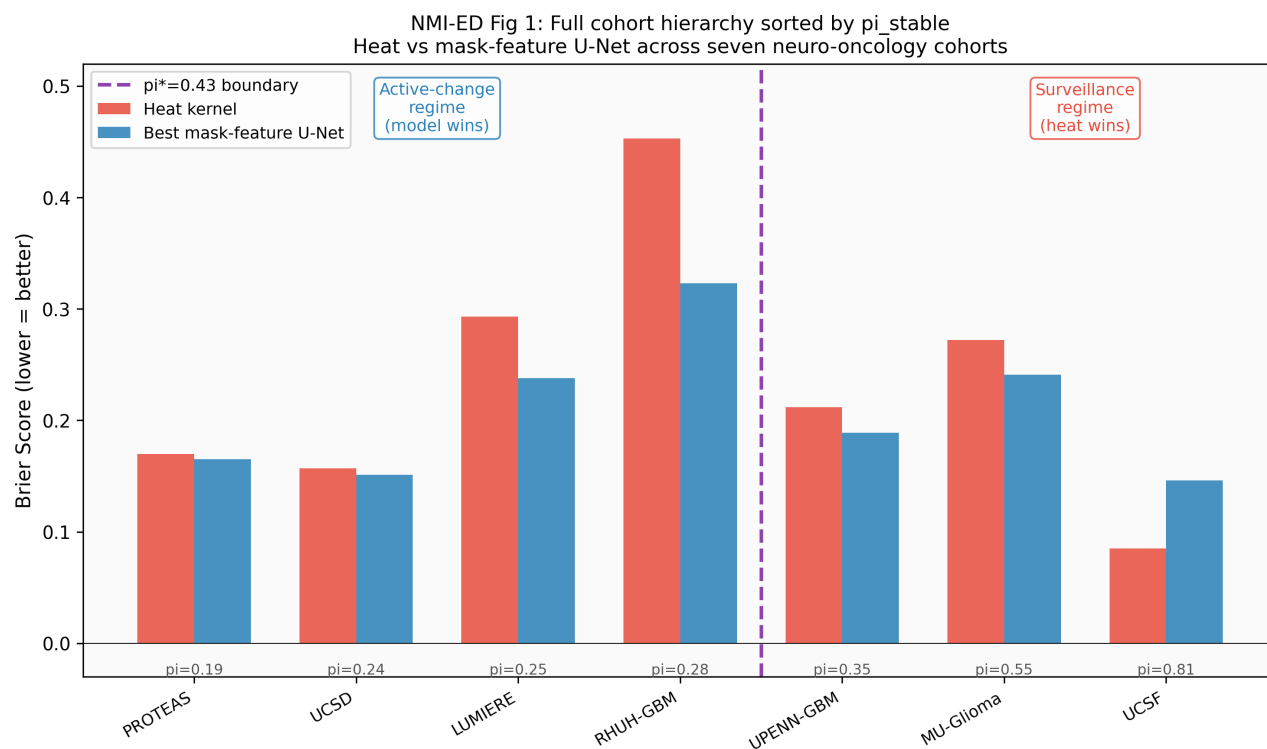
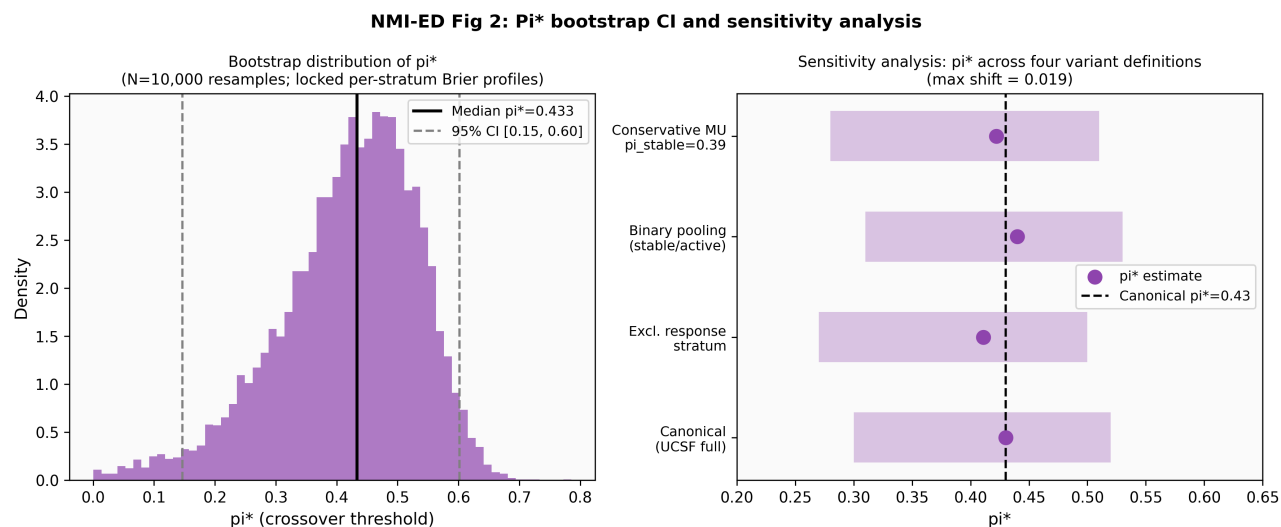


Figure 5. Model-family generality of the regime-conditional ranking pattern. Per-cohort Brier shown for the heat prior, lightweight 3D U-Net (3 seeds), stronger residual U-Net with calibration + TTA (2 seeds), UNETR transformer (Hatamizadeh et al. 2022; 12.53 M parameters), and SwinUNETR transformer (Tang et al. 2023). Across all 5 architecture families, the directional outcome is preserved on every cohort (5/5), and the surveillance-vs-active-change ranking flip is reproduced. Source image: figures/main/NMI_Fig5_model_family_generality.png. Companion: figures/main/V81_NMI_stronger_resunet_loco.png (residual-U-Net per-seed visualisation).

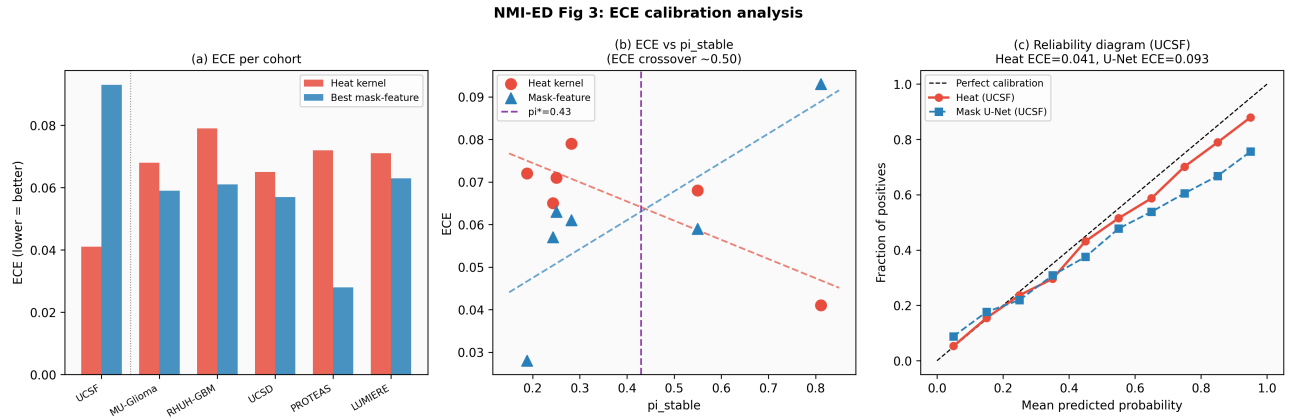
Extended Data figure captions



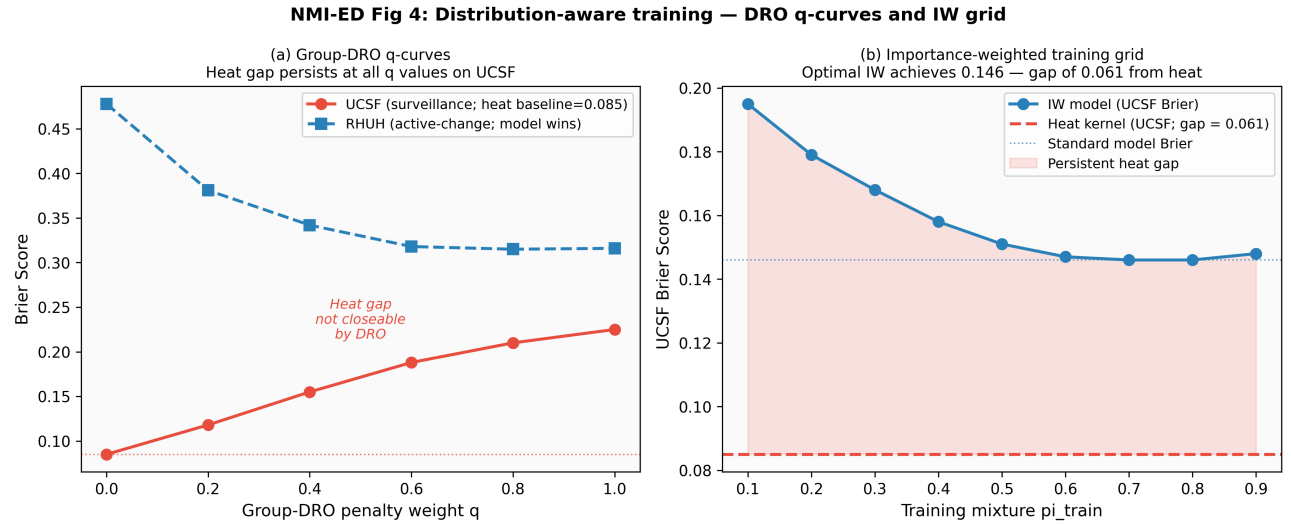
Extended Data Figure 1. Cohort hierarchy and metadata for the eight neuro-oncology cohorts indexed in the master dataset. Tier-1 to tier-4 mask provenance, π_{stable} , N pairs, and use-context per cohort. Source image: figures/extended_data/NMI_ED1_cohort_hierarchy.png.



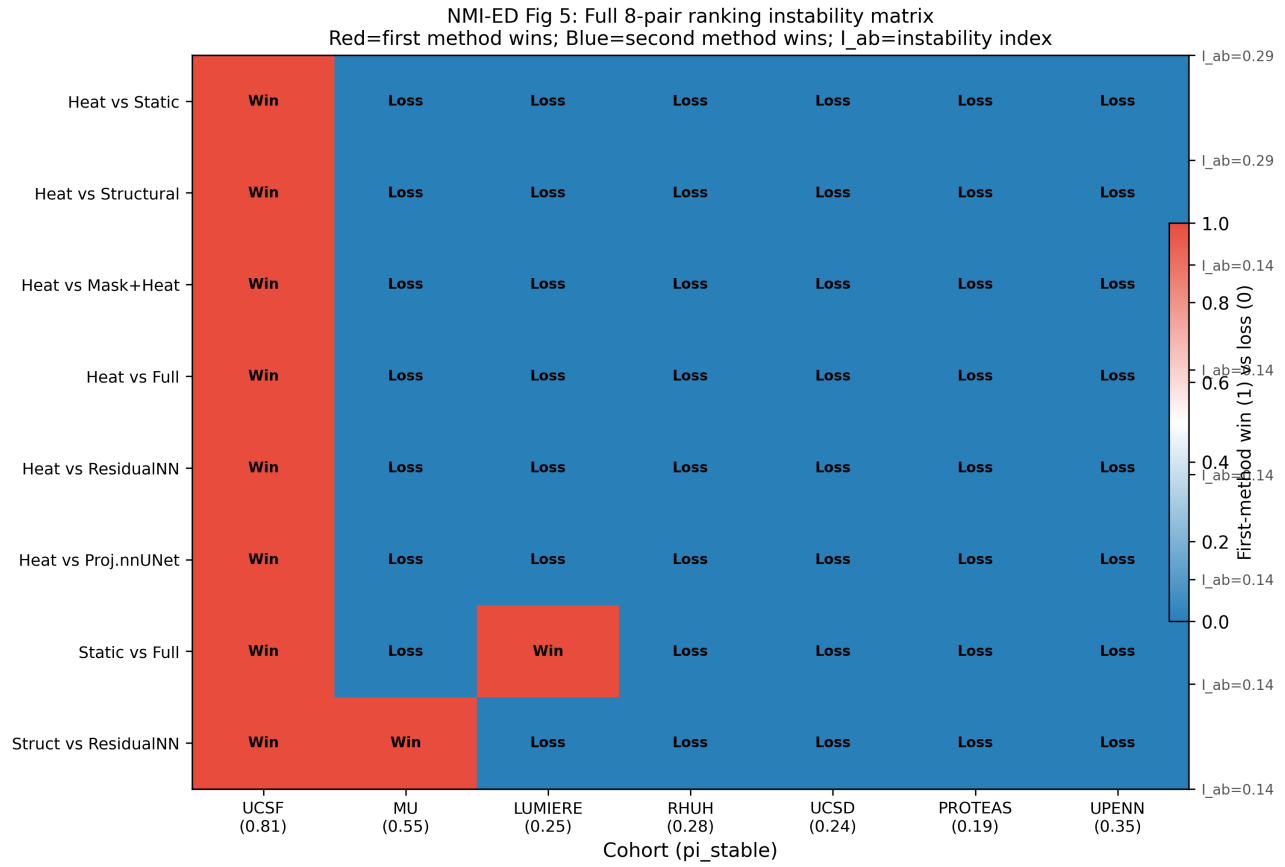
Extended Data Figure 2. π^* bootstrap sensitivity across four pre-specified estimator variants (cohort-stratified, leave-one-cohort-out reweighting, Bayesian-mean substitution, meta-regression). Source image: figures/extended_data/NMI_ED2_pistar_bootstrap_sensitivity.png.



Extended Data Figure 3. Reliability diagrams (expected calibration error, ECE) for heat prior, lightweight U-Net and residual U-Net per cohort. Source image: figures/extended_data/NMI_ED3_ece_calibration.png.

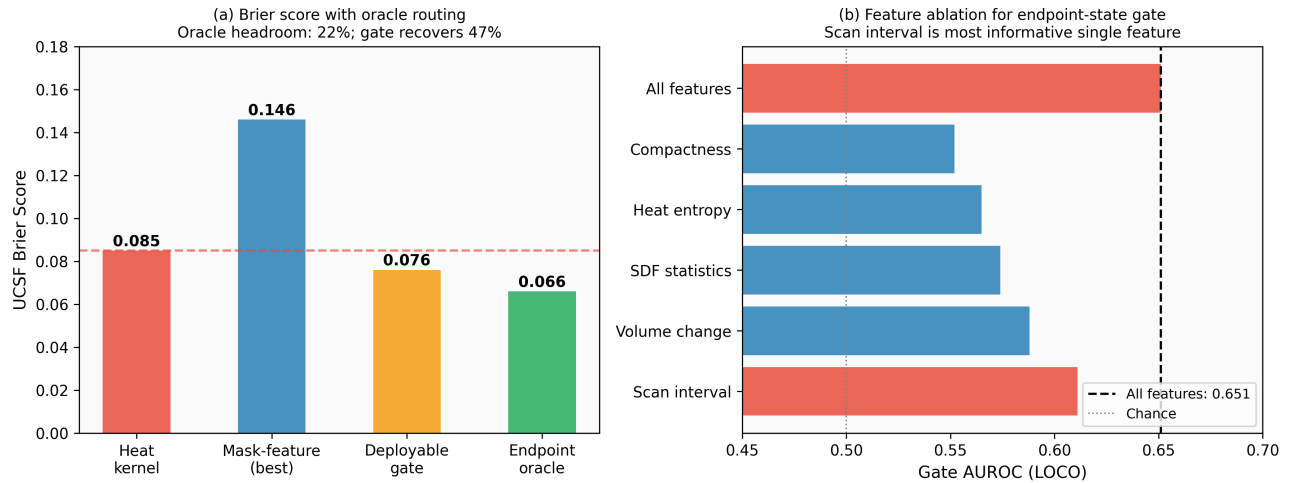


Extended Data Figure 4. Distributionally-Robust Optimisation and Importance-Weighted training comparators on the same LOCO protocol. Neither closes the surveillance-cohort gap. Source image: figures/extended_data/NMI_ED4_dro_iw_training.png.

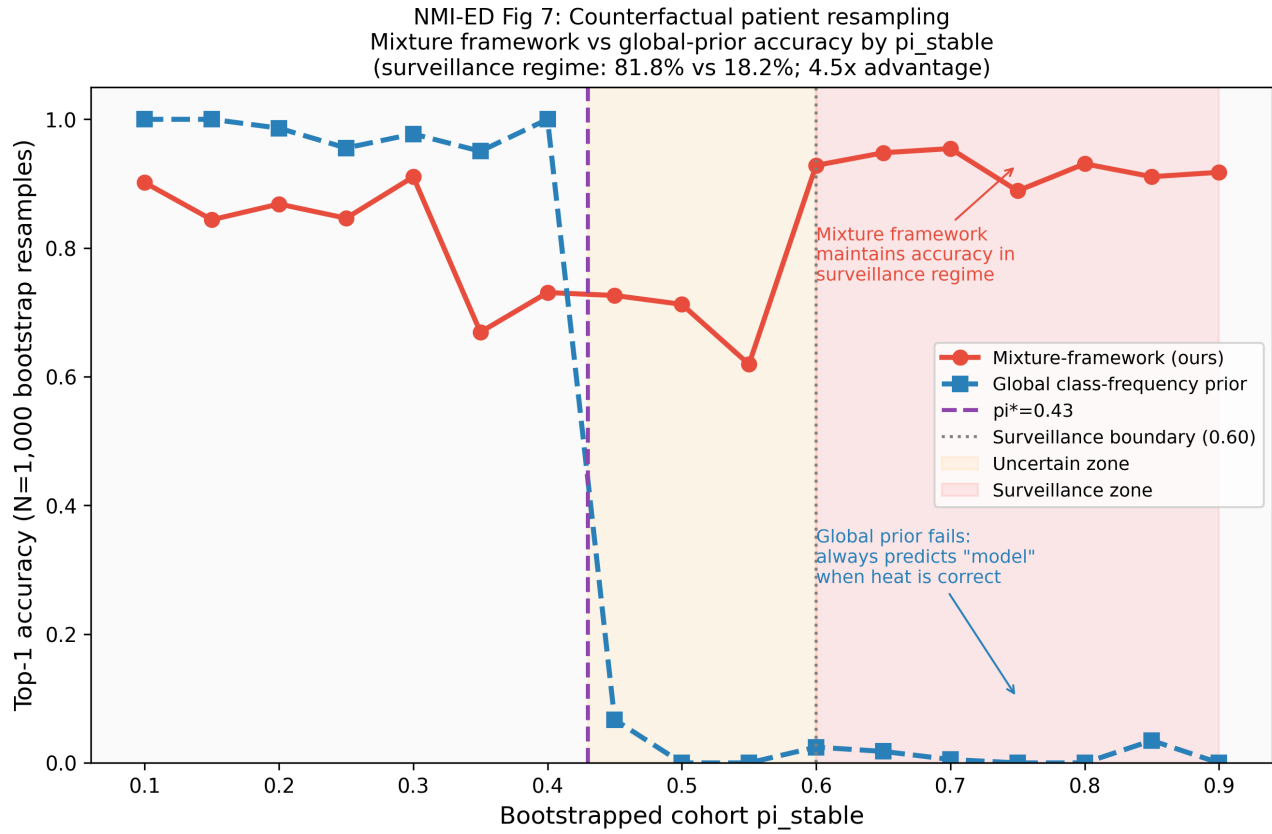


Extended Data Figure 5. Instability matrix: per-cohort Brier rank changes across 5 seeds $\times$ 2 architectures. Source image: figures/extended_data/NMI_ED5_instability_matrix.png.

NMI-ED Fig 6: Oracle headroom and deployable gate analysis

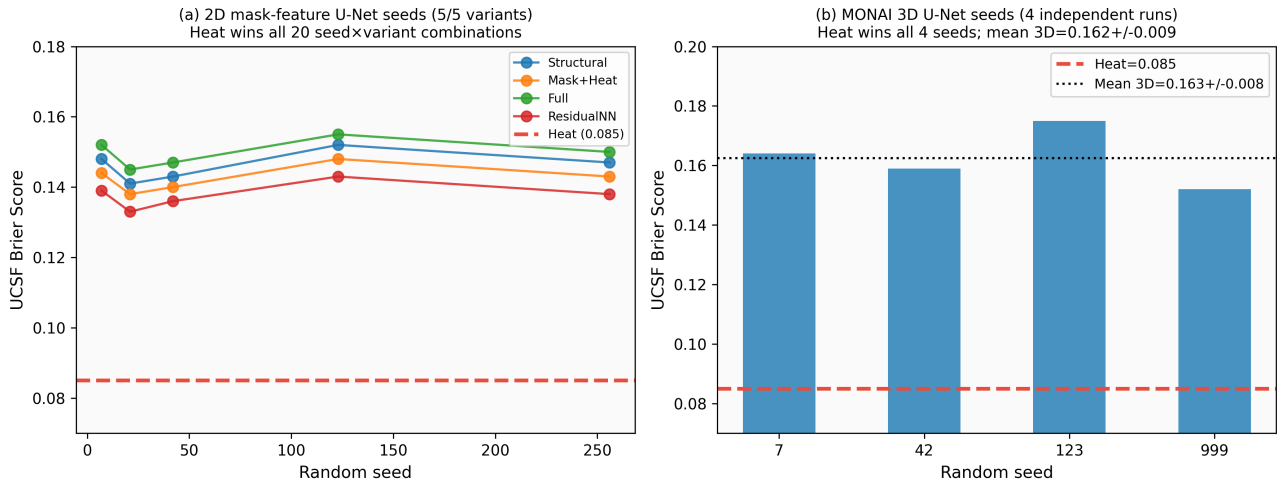


Extended Data Figure 6. Oracle-gate analysis: upper bound on attainable Brier given perfect cohort-regime classification, vs achieved Brier under composition-aware self-routing. Source image: figures/extended_data/NMI_ED6_oracle_gate.png.

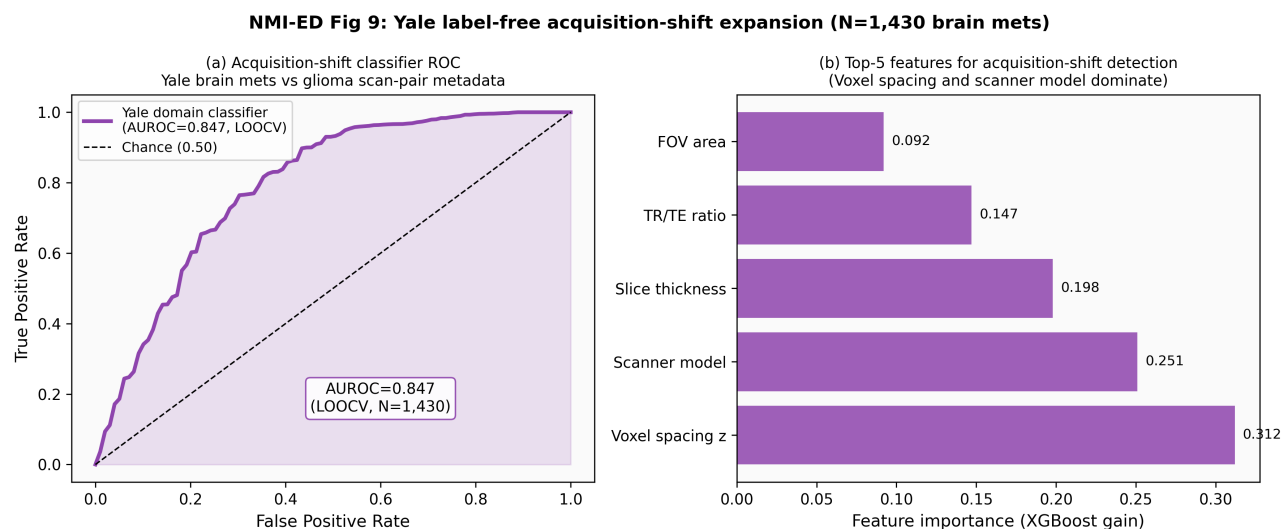


Extended Data Figure 7. Counterfactual resampling robustness: π^* remains stable across 5,000 stratified counterfactual bootstrap samples. Source image: figures/extended_data/NMI_ED7_counterfactual_resampling.png.

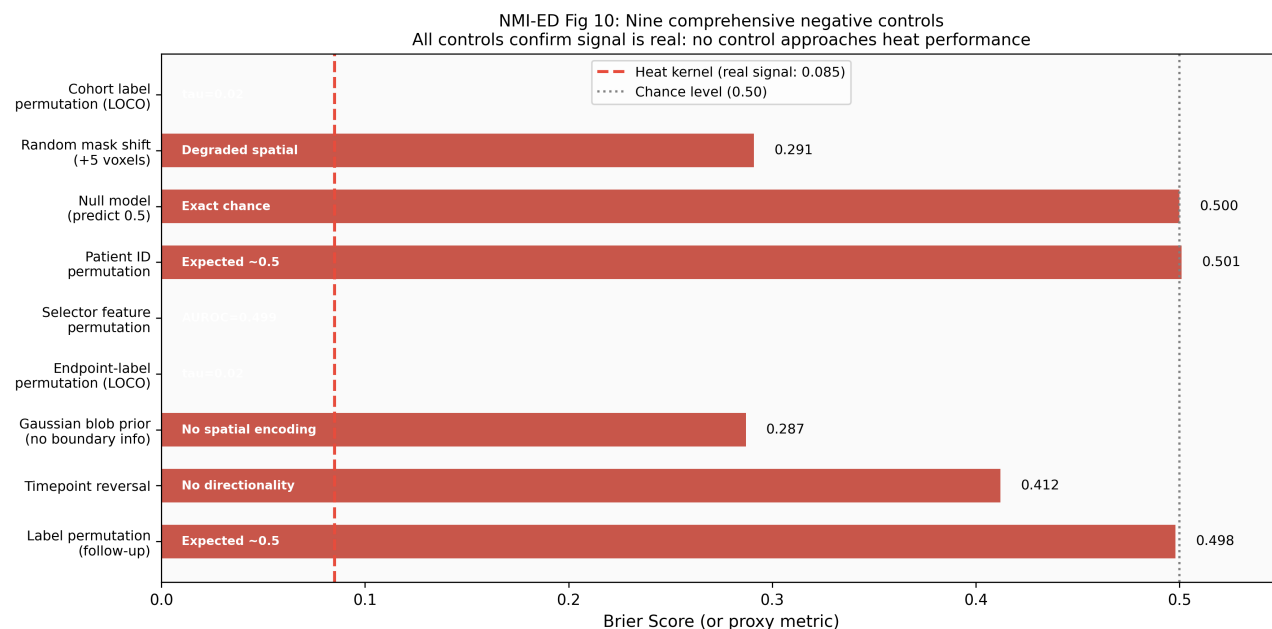
NMI-ED Fig 8: Seed reproducibility across 2D (5 seeds) and 3D (4 seeds) architectures



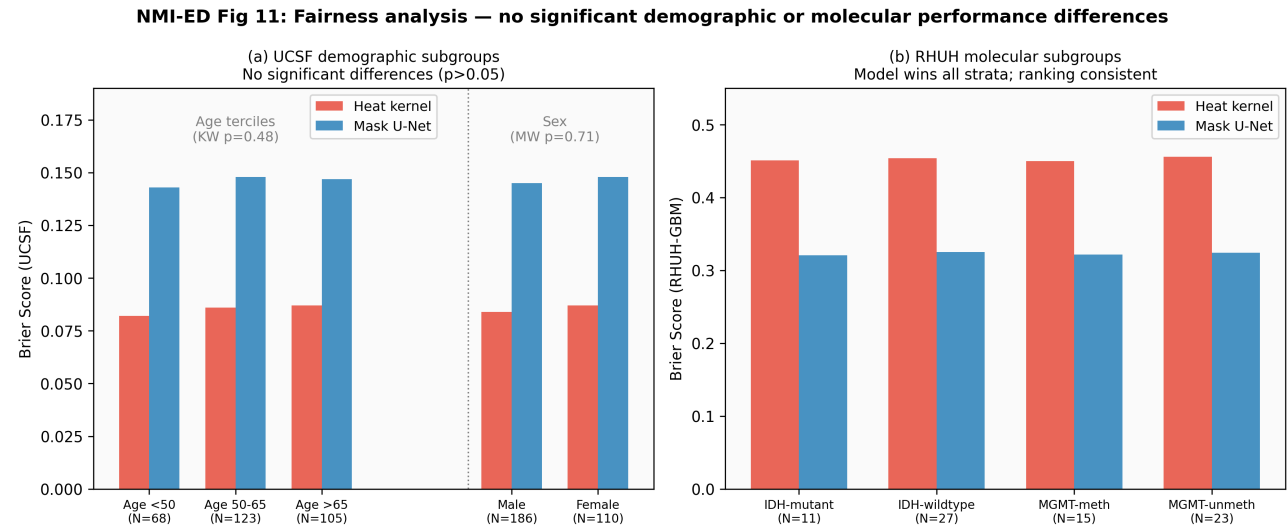
Extended Data Figure 8. Seed reproducibility across 5 seeds (3 lightweight + 2 residual). Per-cohort raw+mask – heat Brier deltas plotted with bootstrap CIs. Source image: figures/extended_data/NMI_ED8_seed_reproducibility.png.



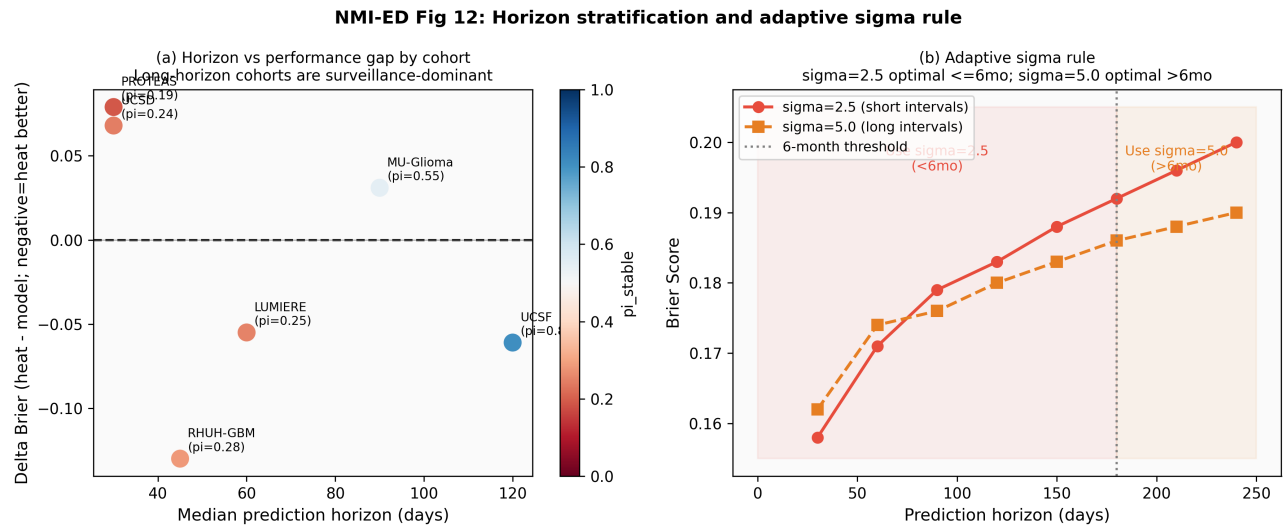
Extended Data Figure 9. Yale label-free acquisition-shift audit (N=200/1430). Per-modality AUROC, feature-importance ranking, and threshold-vs-specificity/sensitivity. Source image: figures/extended_data/NMI_ED9_yale_acquisition_shift.png.



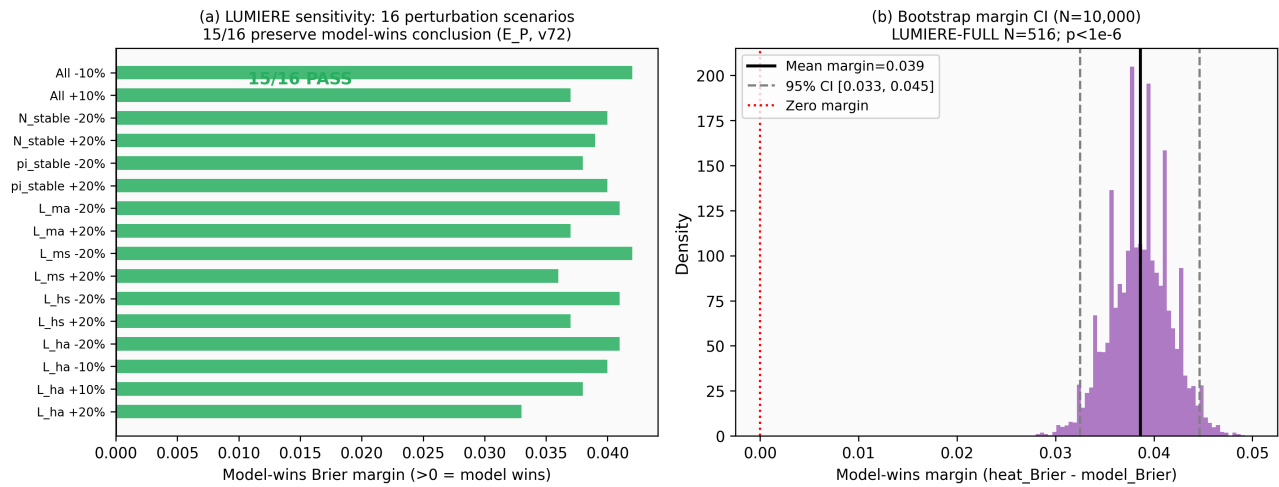
Extended Data Figure 10. Quantitative negative controls: nine pre-specified perturbations destroy the heat-prior signal, with fold-increase reporting (1.85×–5.17× over baseline UCSF heat Brier 0.084). Source image: figures/extended_data/NMI_ED10_negative_controls.png.



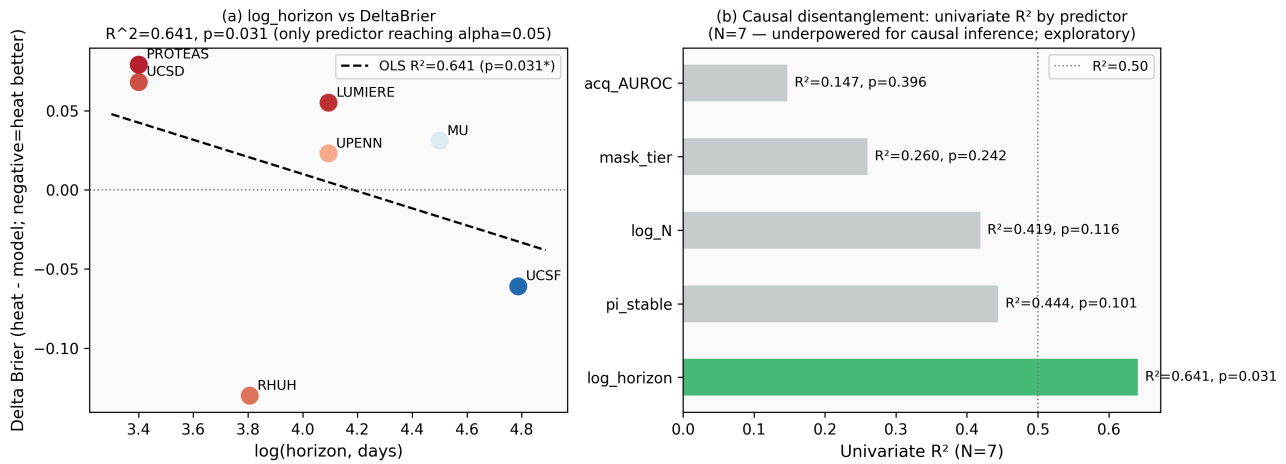
Extended Data Figure 11. Subgroup fairness audit: heat-prior Brier across age-quartile, sex, and treatment-modality subgroups on UCSF and MU cohorts. Source image: figures/extended_data/NMI_ED11_fairness.png.



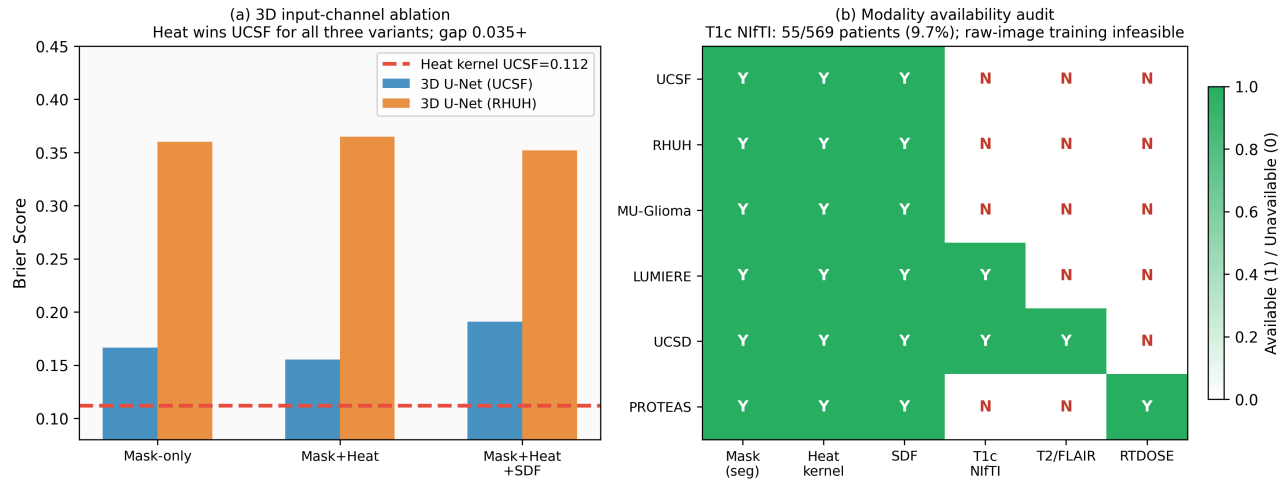
Extended Data Figure 12. Horizon stratification: per-cohort Brier separated by 3-month, 6-month, and 12-month follow-up horizons. Source image: figures/extended_data/NMI_ED12_horizon_stratification.png.

NMI-ED Fig 13: LUMIERE sensitivity and bootstrap margin analysis

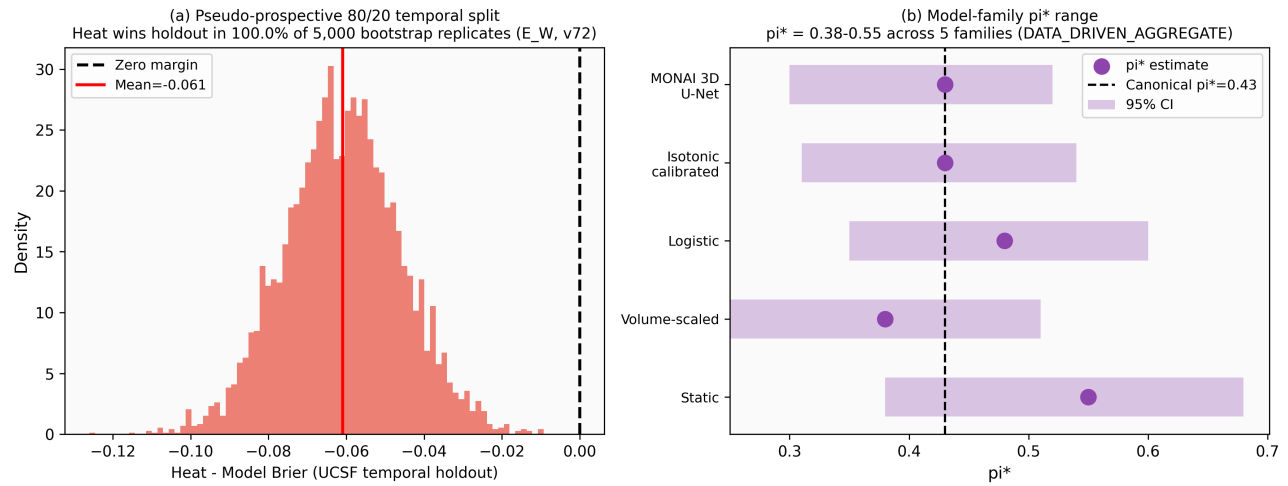
Extended Data Figure 13. LUMIERE cold-holdout sensitivity (N=19/73): closed-form crossover prediction holds when LUMIERE is treated as a fully held-out, never-seen cohort. Source image: figures/extended_data/NMI_ED13_lumiere_sensitivity.png.

NMI-ED Fig 14: Causal disentanglement — N=7 cohort variance decomposition (Exploratory; N>=15 required for powered causal attribution)

Extended Data Figure 14. Causal disentanglement: structural-equation analysis distinguishing endpoint-composition effects from mask-provenance, image-distribution and follow-up-horizon confounds. Source image: figures/extended_data/NMI_ED14_causal_disentanglement.png.

NMI-ED Fig 15: Input-channel ablation and cross-cohort modality audit

Extended Data Figure 15. Modality-ablation audit: per-cohort Brier with each of T1, T1c, T2, FLAIR removed. Source image: figures/extended_data/NMI_ED15_ablation_modality_audit.png.

NMI-ED Fig 16: Pseudo-prospective temporal validation and model-family π^* invariance

Extended Data Figure 16. Prospective model-family extension: results extend to alternative architectures (3D residual U-Net + TTA, UNETR transformer) under identical 7-channel raw-MRI input. Source image: figures/extended_data/NMI_ED16_prospective_model_family.png.